Global identifiability of a behavioral SIR model for every fixed real saturation exponent $k \geq 2$

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Abstract

We prove generic global structural identifiability of the positive real behavioral SIR model for each fixed, known real exponent $k \geq 2$, from exact observation of the infectious population and the known initial total population. This proves a specified formulation of Conjecture 2 of Pant, Saucedo and Pogudin: their conjecture sentence does not explicitly stipulate that the exponent is known or that the domain is positive, whereas these are assumptions of the theorem here. For every fixed exponent, an open dense set of true points is identified against all positive competing points at that same exponent. For rational $k = p/q$, a root substitution gives a two-state rational model whose minimal input–output equation has order two and degree q in its highest derivative. For irrational k, the original supplement's auxiliary-state model has no rational constants beyond the parameter field. A primitive third-order equation and a finite Wronskian then identify the rates and hidden states. Both arguments exclude exceptional exponents throughout $k \geq 2$. The proof is in prose. Separate exact computations check identities and finite cases. The exponent itself is not inferred, and no claim of stable estimation from noisy or discrete observations is made.

1 Model, prior work and result

Consider

$$S' = -\frac{\beta SI}{(1+\alpha I)^k}, \qquad I' = \frac{\beta SI}{(1+\alpha I)^k} - \gamma I, \qquad R' = \gamma I, \qquad y = I. \tag{1}$$

The exponent k and the initial total population $N_0 = S(0) + I(0) + R(0) > 0$ are fixed and known. Parameters α, β, γ and initial states are positive. Powers use the positive real branch. The unknown quantities are the three parameters and the initial states; the state trajectories then follow by uniqueness of the analytic initial-value problem. Structural identifiability is an assertion about exact output germs, not numerical stability.

Pant, Saucedo and Pogudin posed this question in Section 3.6 and Conjecture 2 of [1]. Their Theorem 3.8 gives a two-fold ambiguity at $k = 1$, while Supplement S9 [2] reports global identifiability for $k = 2, 3, 4, 5, 10, 50$. Its Remark 1 already introduces

$$Z = \frac{(1+\alpha I)^k}{\beta} \tag{2}$$

and uses a computed identifiability result for the transformed model and the possible zeros of recovery denominators as functions of k. That argument has no integer restriction but specifies neither a threshold nor all exceptional exponents. The transformation itself is due to that supplement.

Here the rational and irrational cases are proved separately, with no exceptional exponent in the stated range. This proves a specified formulation of Conjecture 2, with each real exponent fixed and known, on the positive real domain with known N_0; the original conjecture sentence does not explicitly give the first two restrictions. The domain choice is substantive: for non-integer rational $k = p/q$, the complex cover identifies β only up to q distinct values (Section 5), whereas the positive branch selects one. The surrounding fixed-exponent examples motivate this formulation, but we do not infer from them a claim about the authors' intended treatment of unknown k. The integer theorem of [5] is included as a

special case on the positive domain. The rational proof uses the identifiability results of Ovchinnikov, Pogudin and Thompson [3] and the model-ideal framework of Hong, Ovchinnikov, Pogudin and Yap [4]. The irrational proof below gives a direct Wronskian argument for the comparison with every positive competitor.

Definition 1. Fix k and $N_0 > 0$. Let Ω_{N_0} be the open set of positive $(\alpha, \beta, \gamma, S_0, I_0)$ with $S_0 + I_0 < N_0$, and put $R_0 = N_0 - S_0 - I_0$. A quantity is generically globally identifiable if there is an open dense set $U \subset \Omega_{N_0}$ such that, for each true point in U, every point of Ω_{N_0} producing the same analytic output germ at time zero has the same value of that quantity. The competing points need not belong to U.

Theorem 2. *For every fixed, known real $k \geq 2$ and every fixed, known $N_0 > 0$, model (1) is generically globally identifiable in the sense of Definition 1. Thus $\alpha, \beta, \gamma, S_0, I_0, R_0$ are determined by $I(t)$ and N_0 at every true point of an open dense set $U_{k,N_0} \subset \Omega_{N_0}$, against all positive competitors at the same exponent and population. The set may depend on k and N_0.*

Present assurance. The theorem has a prose proof and is not formalized in a proof assistant or reviewed by human experts. Separate AI proof checks and separately written exact computations supported the rational and irrational arguments, including their all-competitor quantifiers. The finite calculations in Section 9 check algebraic identities and specified examples; they are not the proof for all real exponents. This extract contains the article TeX and PDF. The companion `README.md` specifies its scope and distinguishes the original supplement and verification record from those of the minimal-jet packet.

2 A rational cover and its constants

In this and the next three sections, write $k = p/q \geq 2$ in lowest terms, with p, q positive integers. Set $K = \mathbb{C}(\alpha, \beta, \gamma)$ and let $L = K(S, z)$, with S, z algebraically independent. Define the output and derivation by

$$y = \frac{z^q - 1}{\alpha}, \qquad DS = -\frac{\beta S y}{z^p}, \qquad Dz = \frac{\alpha}{q} z^{1-q} \left(\frac{\beta S y}{z^p} - \gamma y \right), \qquad D|_K = 0. \tag{3}$$

This is a rational ODE with rational output. A common denominator is $\alpha q z^{p+q-1}$. On the positive real domain the changes of initial coordinates

$$z = (1 + \alpha I)^{1/q} > 0, \qquad I = (z^q - 1)/\alpha$$

are analytic inverses, and $z^p = (1+\alpha I)^k$. Differentiation shows that they carry solutions in both directions. In particular, the cover does not add a free initial condition to the positive real model. The equation for R can be removed, since $R = N_0 - S - I$ and $D(N_0 - S - I) = \gamma I$.

Lemma 3. *The field of constants of (L, D) is exactly K.*

Proof. The derivation $E = (z^p/y)D$ has the same constants as D, and

$$ES = -\beta S, \qquad Ez = \frac{\alpha\beta}{q} S z^{1-q} - \frac{\alpha\gamma}{q} z^{p-q+1}. \tag{4}$$

Equivalently,

$$E = -\beta S \partial_S + \left(\frac{\alpha\beta}{q} S z^{1-q} - \frac{\alpha\gamma}{q} z^{p-q+1} \right) \partial_z.$$

For a nonzero $F \in K(z)(S)$ with $EF = 0$, use the embedding in the formal Laurent series field $K(z)((S))$ and write $F = \sum_{n \geq m} f_n(z) S^n$, $f_m \neq 0$. The derivation does not lower S-order, so it extends continuously to $K(z)((S))$; its term containing $S\partial_z$ raises that order by one. Thus the coefficient of S^m in EF gives

$$-\beta m f_m - \frac{\alpha\gamma}{q} z^{p-q+1} f_m' = 0. \tag{5}$$

If $m \neq 0$, then

$$\frac{f_m'}{f_m} = -\frac{q\beta m}{\alpha\gamma} z^{-(p-q+1)}.$$

The logarithmic derivative of a nonzero rational function has at most a simple pole at $z = 0$: writing $f_m = z^r h$ with h of valuation zero gives $f_m'/f_m = r/z + h'/h$. The right side has pole order $p - q + 1 \geq q + 1 \geq 2$, a contradiction. Hence $m = 0$ and (5) gives $f_0 \in K$. If $F - f_0$ were nonzero, it would again be a constant with positive S-order, giving the same contradiction. Therefore $F = f_0 \in K$. □

3 Differential order two and highest-derivative degree q

Put $u = 1 + \alpha y$, $v = Dy$ and $w = D^2 y$, recalling $k = p/q$. Define

$$A = yuw - \big(1 - (k-1)\alpha y\big)v^2 + k\alpha\gamma y^2 v. \tag{6}$$

The symbol A denotes this polynomial in output derivatives; it is not the auxiliary state Z in (2). From the output equation and conservation we have

$$S = \frac{(v + \gamma y)z^p}{\beta y}, \qquad DS = -(v + \gamma y). \tag{7}$$

Differentiate the first expression and use $D(z^q) = \alpha v$ and the second expression. Multiplication by βy^2 gives

$$(w + \gamma v)z^p y + k\alpha v y(v + \gamma y)z^{p-q} - (v + \gamma y)z^p v = -\beta y^2(v + \gamma y).$$

Using $u - k\alpha y = 1 - (k-1)\alpha y$ and collecting terms gives

$$z^{p-q}A + \beta y^2(v + \gamma y) = 0. \tag{8}$$

Raising this equality to the qth power yields

$$Q(y, v, w) = u^{p-q}A^q - (-\beta)^q y^{2q}(v + \gamma y)^q = 0. \tag{9}$$

Keeping the factor $(-\beta)^q$ makes the sign correct for odd as well as even denominators.

Lemma 4. *The elements y, v are algebraically independent over K. The polynomial Q is irreducible in $K[y, v, w]$ and has minimal degree q in w over $K(y, v)$.*

Proof. Equation (7) gives $K(S, z) = K(v, z)$, so v, z are independent; then y, v are independent as well. Write $F = K(y, v)$. The polynomial $T^q - u$ is Eisenstein at u in $K(v)[u][T]$ (and is linear when $q = 1$), hence $[F(z) : F] = q$. Its conjugates are $z \mapsto \zeta z$ for $\zeta^q = 1$, since $\mathbb{C}$ contains the roots of unity. Equation (8) writes w as an element of F plus a nonzero element of F times $z^{-(p-q)}$. Because $\gcd(p - q, q) = 1$, its q conjugates are distinct. Therefore w has minimal degree q and Q is irreducible in $F[w]$.

The leading coefficient of Q as a polynomial in w is $y^q u^p$. Its content in $K[y, v][w]$ can thus have only the factors y or u. However,

$$Q(0, v, w) = (-1)^q v^{2q}, \qquad Q(-1/\alpha, v, w) = -(-\beta)^q \alpha^{-2q}(v - \gamma/\alpha)^q \tag{10}$$

are nonzero; the second equality uses $p - q > 0$. Thus Q is primitive, and Gauss's lemma proves irreducibility in $K[y, v, w]$. □

Proposition 5. *For the ranking $y < y' < y'' < \cdots$, the set $\{Q\}$ is a monic characteristic presentation of the output ideal of* (3).

Proof. The output ideal J is the kernel of $K\{y\} \to L$ and is prime. This agrees with the output part of the rational-model ideal obtained by saturating by the common denominator: after localization, all state derivatives reduce to rational expressions in the independent states S, z. This is also the model-ideal setting of [4, Lemma 3.2].

The leader of Q is y'', with degree q. Lemma 4 excludes any nonzero element of J of lower rank. An autoreduced set in one differential indeterminate has at most one element; hence $\{Q\}$ is a characteristic set. Its initial $y^q u^p$ belongs to $K[y, v]$, the ring of non-leading variables. Separately, Q itself has no nonconstant factor in $K[y, v]$, by primitivity. Finally $[y^q w^q]Q = 1$. These are the conditions in [3, Definition 5.4]. Here "monic" means that at least one monomial coefficient is one, not that the initial is one. □

4 Coefficients and the integer special case

Proposition 6. *For $q \geq 2$, the field generated over $\mathbb{C}$ by the coefficients of Q is $\mathbb{C}(\alpha, \gamma, \beta^q)$. For $q = 1$ it is K.*

Proof. When $q \geq 2$, direct expansion gives

$$\begin{aligned} [y^q w^q]Q &= 1, & [y^{q+1} w^q]Q &= p\alpha, \\ [y^{q+1} v w^{q-1}]Q &= p\alpha\gamma, & [y^{3q}]Q &= (-1)^{q+1}\beta^q\gamma^q. \end{aligned} \tag{11}$$

For the third coefficient, choose the w term $q-1$ times and the $k\alpha\gamma y^2 v$ term once from A^q, taking the least y power in the remaining factors. Terms containing v^2 cannot contribute. The second summand of (9) has no w, and $q - 1 > 0$. Thus

$$\alpha = \frac{[y^{q+1}w^q]Q}{p}, \quad \gamma = \frac{[y^{q+1}vw^{q-1}]Q}{p\alpha}, \quad \beta^q = \frac{(-1)^{q+1}[y^{3q}]Q}{\gamma^q}. \tag{12}$$

All coefficients lie in this field, so equality follows.

If $q = 1$, $k = p \geq 2$ and $Q = u^{p-1}A + \beta y^2(v + \gamma y)$, which is the integer-exponent polynomial of [5]. In this case the β term contributes to $[y^2 v]Q$, so one instead uses

$$[yw]Q = 1, \quad [y^2 w]Q = p\alpha, \quad [y^3 v]Q = p(p-1)\alpha^2\gamma, \quad [y^2 v]Q = \beta + p\alpha\gamma. \tag{13}$$

These recover α, γ, β successively. The integer factors in the denominators of (12) and (13) never vanish in their respective ranges. The other divisions are legitimate at generic points and on the positive parameter domain. □

5 One experiment and the positive branch

By Proposition 5 and [3, Corollary 5.7], the coefficient field is the field of input–output identifiable parameter functions. Lemma 3 verifies the hypothesis of [3, Theorem 4.7], so it is also the field of functions identifiable from one experiment. In particular, α, γ, β^q are generically globally identifiable on the complex rational cover.

The quantifiers here are those of [3, Notation 2.1 and Definition 2.2], which extend the definition cited there from [4]. For this autonomous cover, let $\Omega_c = \{(\alpha, \beta, \gamma, S_0, z_0) : \alpha q z_0^{p+q-1} \neq 0\}$. For each identifiable function h, the true point lies in a nonempty Zariski open set $\Theta \subset \Omega_c$, while every competing point in Ω_c where h is defined and the output germ is the same must give the same value of h. Competitors are not restricted to Θ. For $h = \alpha, \gamma, \beta^q$ these functions are defined everywhere, and we intersect their three Zariski open sets. This is the quantifier structure specialized to the positive branch in Definition 1.

For $q > 1$, β itself is not globally identifiable on that cover. For each $\zeta^q = 1$, the transformation

$$(z, \beta, S, \alpha, \gamma) \longmapsto (\zeta z, \zeta^p \beta, S, \alpha, \gamma) \tag{14}$$

preserves the output and DS and multiplies Dz by ζ. It therefore takes solutions to solutions. Coprimality gives q distinct generic values of β, and identifiability of β^q allows no more than q.

Rational case of Theorem 2. Take a true point in the Zariski open set Θ just described for the complex-cover identifiability statement, and any positive real competing point with the same output. Both lift by (3). The global statement compares the true point with all admissible competitors, so α, γ, β^q agree. Their positive roots $z = (1 + \alpha y)^{1/q}$ agree as well. The relation before taking powers gives

$$\beta = -\frac{z^{p-q}A}{y^2(v + \gamma y)}. \tag{15}$$

This recovers the same β for both solutions. Its denominator is nonzero when $\beta S y z \neq 0$, since $v + \gamma y = \beta S y / z^p$. For positive β one could alternatively take the positive root of β^q; equation (15) displays the role of the branch.

Combining (7) with (8) gives

$$S = -\frac{yu(v + \gamma y)^2}{A}, \qquad I = y, \qquad R = N_0 - S - I. \tag{16}$$

The right sides coincide. In fact, along every positive solution, (8) gives $z^{p-q}A = -\beta^2 S y^3 / z^p \neq 0$. Thus both solutions have the same nonzero denominator, proving equality of all initial states. The rational expression for S also gives its identifiability on the cover by [3, Proposition 3.1].

For each fixed p, q, intersect Θ with the nonvanishing sets of the displayed denominators. This is a nonempty Zariski open set, by the cited definition and finite intersection. The positive real cover with $S + I < N_0$ is a nonempty Euclidean open set in real coordinates. A nonzero complex polynomial cannot vanish on that whole open set, so the intersection is open and dense there. The analytic coordinate change in Section 2 transfers it to Ω_{N_0}. This restricts the true point only; it does not restrict the competing points to a generic subset. □

For $q = 1$, no root choice is needed. The same coefficient and state formulas recover the complex generic integer result of [5]. The new real-branch theorem therefore includes its positive real restriction and does not assert a contrary ambiguity for integers.

6 The irrational auxiliary-state field

Now fix an irrational real $k \geq 2$, and use the state Z of (2). In this section and the next two, redefine $K = \mathbb{C}(\alpha, \gamma)$, in contrast to $K = \mathbb{C}(\alpha, \beta, \gamma)$ in the rational case: β is encoded by $Z(0)$, not included as an independent parameter of this field. Set $L = K(S, u, Z)$ with three algebraically independent states. The output and derivation are

$$\begin{aligned} y &= (u-1)/\alpha, & DS &= -Sy/Z, \\ Du &= \alpha y(S/Z - \gamma), & DZ &= k\alpha y(S - \gamma Z)/u. \end{aligned} \tag{17}$$

The parameter β does not occur in the vector field: it is encoded by the initial value $Z(0) = u(0)^k/\beta$. This is the model of S9, with a changed name for its auxiliary state.

Lemma 7. *For each fixed irrational $k \geq 2$, the field of constants is $L^D := \{H \in L : DH = 0\} = K$.*

Proof. Rescale by the nonzero field element Z/y and set $B = Z/u$. The derivation $E = (Z/y)D$ has the same constants and is

$$E = -S\partial_S - \alpha\gamma BX + \alpha S\left(\partial_u + \frac{(k-1)B}{u}\partial_B\right), \qquad X = u\partial_u + (k-1)B\partial_B. \tag{18}$$

It does not lower S-order and extends continuously to $K(u, B)((S))$. For a nonzero rational H with $EH = 0$, write $H = \sum_{n \geq m} h_n(u, B)S^n$, $h_m \neq 0$. Its leading coefficient satisfies

$$E_0 h_m = m h_m, \qquad E_0 = -\alpha\gamma BX.$$

Expand $h_m = \sum_{j \geq n} g_j(u)B^j$ in $K(u)((B))$. Since

$$E_0(g_j B^j) = -\alpha\gamma\big(ug_j' + (k-1)jg_j\big)B^{j+1},$$

E_0 raises the lowest B-order by at least one. The equality is impossible for $m \neq 0$. Thus $m = 0$ and $Xh_0 = 0$.

Write $h_0 = P_0/Q_0$ with coprime polynomials in $K[u, B]$. The equality $Q_0 X P_0 = P_0 X Q_0$ implies that P_0 divides XP_0 and Q_0 divides XQ_0. Since X does not increase total degree, $XP_0 = cP_0$ and $XQ_0 = cQ_0$ for the same $c \in K$. The weights $i + (k-1)j$ of distinct monomials $u^i B^j$ are distinct: a coincidence with different j would make k rational, and equal j forces equal i. Each eigenpolynomial is therefore a monomial. The common eigenvalue gives the same exponents, so $h_0 \in K$. If $H - h_0$ were nonzero, it would be an E-constant of positive S-order, contradicting the preceding argument. Hence $H \in K$. □

The irrationality is used only in the monomial weights. For rational $k = p/q$, the three-state field instead has the nonconstant rational first integral u^p/Z^q. This does not contradict Lemma 3: on the two-state cover that expression is β^q, in the parameter field. For every real k, the positive model also has the analytic first integral Z/u^k. In the irrational case it is not rational. Indeed, a putative polynomial relation for u^k over $\mathbb{C}(u)$ gives a finite linear combination of distinct powers u^{i+kj}; after $u = e^s$, a Vandermonde argument for exponentials forces every coefficient to vanish. The lemma concerns rational constants, not all analytic first integrals.

7 Third-order equation and coefficient recovery

Use the jet variables $v = Dy$, $w = D^2y$ and $x = D^3y$. Retain the polynomial A from (6), now with the fixed irrational value of k, and put $r = v + \gamma y$ and $u = 1 + \alpha y$. The original equations give $S = Zr/y$, $DS = -r$ and $DZ/Z = k\alpha v/u$. Differentiating the first identity and multiplying by y^2u gives

$$Z\{k\alpha vyr + uy(w + \gamma v) - urv\} = -y^2ur.$$

The expression in braces is A, so

$$ZA = -y^2ur, \qquad Z = -\frac{y^2ur}{A}, \qquad S = -\frac{yur^2}{A}, \qquad K(S, u, Z) = K(y, v, w). \tag{19}$$

In particular y, v, w are algebraically independent. Logarithmic differentiation of Z yields

$$\frac{DA}{A} = \frac{2v}{y} + \frac{w + \gamma v}{r} - (k - 1)\frac{\alpha v}{u}.$$

Define

$$\begin{aligned} P = {} & uyr\,(v\partial_y A + w\partial_v A + x\partial_w A) \\ & - A\{2uvr + uy(w + \gamma v) - (k - 1)\alpha vyr\}. \end{aligned} \tag{20}$$

It vanishes in L and is linear in x, with

$$\partial_x P = y^2u^2r. \tag{21}$$

Lemma 8. *For each fixed $k \neq 0, 1$, the polynomial P is primitive and irreducible in $K[y, v, w, x]$. Its differential order is minimally three and its degree in the highest derivative is minimally one. Its coefficients generate $K = \mathbb{C}(\alpha, \gamma)$.*

Proof. The content in $K[y, v, w][x]$ can have only the factors y, u, r. They are excluded by

$$\begin{aligned} P|_{y=0} &= 2v^4, \\ P|_{y=-1/\alpha} &= k(k - 1)v^2(v - \gamma/\alpha)^2, \\ P|_{v=-\gamma y} &= -y^2u^2(w - \gamma^2 y)^2. \end{aligned} \tag{22}$$

These are nonzero for $k \neq 0, 1$. Primitivity and linearity in x give irreducibility by Gauss's lemma. The field equality in (19) proves that no lower-order relation exists; linearity gives the minimal positive degree. Moreover the kernel of $K[y, v, w, x] \to L$ is exactly (P): over $K(y, v, w)$ it is generated by the linear equation, and primitivity contracts this ideal to (P). The highest-derivative terms give

$$[y^2vx]P = 1, \qquad [y^3vx]P = 2\alpha, \qquad [y^3x]P = \gamma. \tag{23}$$

All coefficients belong to K, and these recover its generators. □

For the differential ranking by order, $\{P\}$ is also a monic characteristic presentation: it has minimal rank in the prime output ideal, its initial is in the non-leading-variable ring, it has no nonconstant factor there, and a monomial coefficient is one. These are the same conditions as in Proposition 5. One may therefore use [3, Corollary 5.7 and Theorem 4.7] with Lemma 7. The following direct argument spells out why only the true point needs to be generic.

8 Wronskians and all positive competitors

We use the characteristic-zero Wronskian criterion: elements of a differential field are linearly independent over its constants if and only if their Wronskian is nonzero. For completeness, divide columns by a nonzero first element. Elementary row operations give $W(f_1, \dots, f_n) = f_1^n W(1, f_2/f_1, \dots, f_n/f_1)$; expansion along the first column reduces to the Wronskian of $D(f_2/f_1), \dots, D(f_n/f_1)$. If it vanishes, induction gives a nontrivial constant linear combination of those derivatives equal to zero. Its antiderivative is a constant, giving a relation among the original elements. The converse follows by differentiating a constant linear relation. The one-element case starts the induction.

For the fixed irrational k, let $\mathcal{M}$ be the union, as α, γ vary, of the output-monomial supports of P. Expansion of (20) gives $|\mathcal{M}| = 25$ for irrational k, so the Wronskian below has $n = 24$ columns. Some coefficients vanish at rational exponents (for example, $k = 2$), but this section fixes an irrational exponent. All monomials have total degree at most six. The polynomial P has total degree exactly six, as seen from the term $\alpha^2\gamma y^5 x$ in (21). A K-linear relation among the evaluations of $\mathcal{M}$ in L is a polynomial in the kernel (P) of Lemma 8. Its degree is at most six, so it is a scalar multiple of P. There is exactly one relation up to scalar.

Remove $m_0 = y^2 vx$, whose coefficient in P is one. The remaining monomials $m_1, \ldots, m_n$ are K-linearly independent in L. By Lemma 7 and the Wronskian criterion,

$$W = \det\big(D^{i-1} m_j\big)_{1 \le i,j \le n} \neq 0 \quad \text{in } K(S, u, Z). \tag{24}$$

This proves nonvanishing as a rational function; a numerical Wronskian calculation is not needed.

Restrict the true initial point so that W is defined and nonzero. Include the common denominator $\alpha u Z$ and the denominators used to write W; their nonvanishing defines a nonempty Zariski open set in the transformed parameter-state space. Any positive competitor with the same output obeys its own polynomial P_*, with parameters α_*, γ_* but the same fixed k. Its support is contained in $\mathcal{M}$, and its coefficient of m_0 is one. Thus $P_* - P$ is a constant linear combination of $m_1, \ldots, m_n$ along the true output. Differentiate this zero relation $0, \ldots, n-1$ times and evaluate at time zero. The coefficient differences lie in the kernel of the invertible Wronskian matrix, so all are zero. Equation (23) gives $\alpha_* = \alpha$ and $\gamma_* = \gamma$. No condition on the competitor's own Wronskian was imposed.

Irrational case and completion of Theorem 2. The positive initial-coordinate map

$$(\alpha, \beta, \gamma, S_0, I_0) \longmapsto (\alpha, \gamma, S_0, u_0 = 1 + \alpha I_0, Z_0 = u_0^k/\beta) \tag{25}$$

is analytically invertible, with inverse $I_0 = (u_0 - 1)/\alpha$ and $\beta = u_0^k/Z_0$. It carries positive solutions of (1) to (17) and back: differentiating u^k/Z along the latter system gives zero. The image domain is the nonempty real open set $\alpha, \gamma, S_0, Z_0 > 0$, $u_0 > 1$ and $S_0 + (u_0 - 1)/\alpha < N_0$. A nonzero complex polynomial cannot vanish on that entire open set. Hence the Wronskian's nonvanishing set is open dense there, and its inverse image under (25) is open dense in Ω_{N_0}.

For a true point there and any positive competitor, the preceding argument gives the same α, γ. Equation (19) then gives the same S, Z, u from the common output jets. It is defined at every positive competitor, since $y > 0$, $u > 1$, $r = Sy/Z > 0$ and $A = -y^2 ur/Z < 0$. Finally

$$\beta = \frac{u_0^k}{Z_0}, \qquad I_0 = y(0), \qquad R_0 = N_0 - S_0 - I_0$$

agree. Together with the rational case, this proves the theorem for every real $k \ge 2$. No limiting argument from rational exponents is used. □

9 Exact finite checks

For rational exponents, two sets of computations supply 22 case records at 19 distinct exponents, 18 of them in the theorem's range. The producer side consists of a SymPy program for symbolic rational-function identities and a separate standard-library program for sparse polynomials and finite-field Taylor matrices. It covers

$$\frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2}, \frac{11}{2}, \frac{13}{2}, \frac{15}{2}, \frac{17}{2}, \frac{19}{2}, \frac{7}{3}, \frac{8}{3}, \frac{11}{4}, 2, 3.$$

For each case, it checks (8)–(9), the coefficient recovery, primitivity substitutions, (4), (14) and (16). The second set uses separately written sparse Laurent polynomials over $\mathbb{Q}$, with parameters left symbolic. It covers

$$\frac{13}{6}, \frac{17}{5}, \frac{23}{7}, \frac{9}{4}, \frac{31}{8}, 2, 3, \frac{5}{2}$$

and checks the same algebraic identities and selected coefficients. The three overlapping exponents are regression cases. The value $3/2$ is a comparison below the theorem's range, not an instance of Theorem 2.

Only the first 14 cases have finite-field observation matrices. At $(\alpha, \beta, \gamma, S, z) = (2, 3, 5, 7, 2)$, Taylor series are computed exactly modulo 1000003. If m is the number of output monomials in Q, their series

are retained through degree $m+5$ (state series through $m+7$). All denominators are invertible modulo this prime. The column for $y^q w^q$, whose coefficient is one, is removed; the remaining $m-1$ columns have full rank in every case. For $k = 5/2$ the rank is 31, and for $k = 11/4$ it is 161. The matrices and nonzero minors belong to the original companion supplement, outside this article-only extract (see its `README.md`).

A nonzero minor at this point modulo the prime certifies that the corresponding rational-function minor is not identically zero. On its generic nonvanishing set, any two competing coefficient vectors lie in the kernel of the same observation matrix and share the normalized coefficient one. Their difference has that coefficient zero; independence of the other columns makes the difference zero. This certifies generic coefficient uniqueness for these fixed exponents. It is distinct from extrapolating a finite list to all rational k; that conclusion rests on the preceding proof.

For irrational exponents, two producer programs evaluate the three-state identities using sparse Laurent arithmetic over $\mathbb{Q}$ and SymPy, respectively. Their full expansions of P agree in all 36 terms (25 output monomials). They check the original ODE substitution, both recovery identities, the three content substitutions, and the coefficients in (23). The CAS path specializes symbolically at $\sqrt{5}$, π and $1+\sqrt{2}$, without replacing these values by approximate rationals. The constants-field transformations were also checked on 27 Laurent monomials, totaling 90 expanded coefficient terms, with 81 weight pairs $(i,j) \in \{-4,\dots,4\}^2$ and rational controls; the all-rational-function conclusion remains the prose lemma.

A separately written standard-library implementation derives state jets by formal Taylor integration of (17), agrees on all 36 polynomial terms, and works in $\mathbb{Q}(\sqrt{7})$ at $(\alpha,\gamma,S_0,I_0,Z_0) = (2,3,5,2,4)$, $(2,3,12,2,4)$ and $(1,1,7,3,2)$. It checks the equation and both recovery identities through Taylor order eight. The middle case has $y'(0) = 0$ and $y''(0) = -3$, so it includes a local infection maximum. With $N_0 = 20$ all three cases have positive R_0. The original companion supplement, outside this extract, retains a rejected missing-factor formula as a negative control: at the maximum its initial residual vanishes but the next Taylor coefficient does not. This shows why checking a single value is insufficient. Neither these examples nor the 27 monomial tests are a computation of the general Wronskian in (24).

10 Scope of the conclusion

The theorem covers all real exponents $k \geq 2$, each fixed and known, on the positive domain. It neither compares different unknown exponents nor asserts a common generic set for all k. The generic set in the original coordinates is open dense; Zariski openness in those coordinates is not claimed. In the irrational case the initial coordinate change uses u^k and is analytic rather than algebraic. Unknown N_0, identifiability at every initial point, and stable recovery from finite noisy observations remain outside the result.

The rational proof in fact only requires $p > q$, hence also applies to fixed rational $1 < k < 2$: the logarithmic-derivative pole has order $p-q+1 \geq 2$, the content calculation uses $p-q > 0$, and the recovery coefficients remain valid; $q = 1$ then already means $p \geq 2$. The irrational argument likewise uses only irrationality and $k \neq 0, 1$; it remains valid for every irrational real k, with the exclusions automatic for irrational k. These observations are scope notes, not extensions of the main theorem's stated range $k \geq 2$.

The auxiliary-state construction and large-exponent argument of S9 are prior work. The contribution here is the explicit treatment of the constants, minimal equations, exceptional exponents and positive competitors across the stated range. A bounded literature comparison found no matching all-real-exponent theorem in the sources examined; this does not establish priority or exclude other formulations known to specialists.

AI-use statement

AI systems assisted with proof development, separate checks of the proof and algebra, literature organization, computation and manuscript drafting. The author is responsible for the mathematical claims and their stated scope; AI systems are not authors.

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